

SOC Internship Program – Day 26

Threat Intelligence Lifecycle

1. What is Threat Intelligence?

Threat Intelligence is the collection and analysis of information about cyber threats, attacker activities, and malicious indicators. It helps Security Operations Center (SOC) teams identify, detect, and respond to cyberattacks effectively.

2. Importance of Threat Intelligence

Threat intelligence improves cybersecurity operations by helping organizations:

- Detect known threats quickly
- Improve SIEM detection capabilities
- Understand attacker tactics and behavior
- Strengthen incident response processes

3. Threat Intelligence Lifecycle

The Threat Intelligence Lifecycle is a structured process used to gather and use intelligence efficiently.

Stages of the Lifecycle

1. **Planning**
Define security goals and identify intelligence requirements.
2. **Collection**
Gather threat data from multiple sources such as logs, threat feeds, dark web monitoring, and OSINT.
3. **Processing**
Convert raw data into a usable format for analysis.
4. **Analysis**
Study the data to identify threats, patterns, and attacker techniques.
5. **Dissemination**
Share intelligence with SOC analysts, security teams, and management.
6. **Feedback**
Improve the process based on team feedback and incident outcomes.

4. Types of Threat Intelligence

Strategic Intelligence

High-level information focused on business risks and long-term security planning.

Tactical Intelligence

Information about attacker tactics, techniques, and procedures (TTPs).

Operational Intelligence

Details about specific attacks, campaigns, or threat actors.

Technical Intelligence

Technical indicators such as malicious IPs, domains, and file hashes.

5. Common Indicators of Compromise (IOCs)

Common IOCs used in SOC investigations include:

- Malicious IP addresses
- Suspicious domains
- File hashes
- Malware signatures

6. Example Investigation

Scenario

An alert is triggered because a system connects to a malicious IP address.

Threat Feed Result

The IP address is linked to ransomware activity.

SOC Analysis

The SOC team investigates the affected host and identifies a possible compromise. Immediate containment and response actions are initiated.

7. SOC Workflow

Threat Feed

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SIEM Enrichment

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IOC Match

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SOC Investigation

8. SOC Team Roles

L1 Analyst

- Monitor IOC alerts
- Perform initial triage

L2 Analyst

- Investigate intelligence matches
- Analyze suspicious activity

L3 Analyst

- Improve SIEM and threat intelligence integrations
- Develop advanced detection strategies

Assignment Answers

1. Explain Threat Intelligence

Threat intelligence is information related to cyber threats, attacker behavior, and indicators of compromise that helps organizations improve threat detection, prevention, and incident response.

2. Describe Lifecycle Stages

The lifecycle stages are:

- Planning
- Collection
- Processing
- Analysis
- Dissemination
- Feedback

These stages help SOC teams collect and use threat intelligence effectively.

3. Analyze Malicious IP Scenario

If a device connects to a malicious IP linked to ransomware activity, the SOC team should:

- Investigate the affected system
- Check SIEM logs for suspicious behavior
- Isolate compromised devices
- Scan for malware or ransomware indicators
- Block the malicious IP across the network
- Start incident response procedures